

CAPSTONE PROJECT PROPOSAL COHORT-6 BATCH-4

Meeting Intelligence & Action Tracking System

1. Project Title

Meeting Intelligence & Action Tracking System

2. Brief Problem Statement

Meeting recordings contain important information such as discussion topics, decisions, responsibilities, deadlines, risks, and follow-up requirements. Manually listening to recordings, preparing meeting notes, identifying action items, and distributing the resulting information is time-consuming and can lead to missed or inconsistently recorded tasks.

The proposed Meeting Intelligence & Action Tracking System automates this process. A user uploads a meeting recording through an intake form. The workflow sends the recording to AssemblyAI for transcription and speaker diarization. The resulting speaker-aware transcript is passed to an OpenAI model, which extracts a structured meeting summary, key topics, decisions, action items, assignees, deadlines, priorities, risks, follow-up requirements, and parking-lot items. The structured output is then used to create Jira tasks, generate meeting notes, email participants, and create a follow-up calendar event when required.

3. Background Information

3.1 Domain Information

The project combines Generative AI, Natural Language Processing, speech-to-text processing, speaker diarization, structured information extraction, and workflow automation. The current implementation uses n8n as the orchestration layer, AssemblyAI for speech processing, and OpenAI for meeting-intelligence extraction.

3.2 Problem Description and Analysis

Meeting platforms such as Zoom, Microsoft Teams, and Google Meet now offer transcription, AI summaries, and action-item extraction with tentative owners. Their value chain ends there: the output is a static document, and converting it into tracked work, scheduled follow-ups, archived records, and verified completion is left to the participants. This "last mile" is where most meeting outcomes are lost.

This capstone project treats extracted action items as the starting point rather than the final deliverable. An intake form captures the meeting title, participant emails, and a recording (MP3, M4A, WAV, MP4, JSON, or WEBM). The recording is transcribed via AssemblyAI with speaker labels, polled to completion, and assembled into a diarized transcript. An OpenAI extraction step then returns structured JSON containing the summary, decisions, action items, risks, follow-up meeting details, and parking-lot items, with the model explicitly instructed to use "Unassigned" or "No deadline" rather than invent assignees or dates.

Where existing platforms stop, the system continues with four downstream integrations:

1. **Task creation** — action items become tickets in a Jira-style platform with owner, deadline, and meeting reference.
2. **Follow-up scheduling** — identified follow-ups generate calendar invitations for the relevant participants.
3. **Document archival** — transcript, summary, and structured output are stored in Google Drive.

4. **Completion tracking** — tasks are monitored against deadlines, surfacing overdue or unassigned items.

Together, these turn meeting intelligence from passive summarisation into an active accountability mechanism that current platforms leave to manual effort.

3.3 Possible Applications

Below are possible applications

- Corporate and project team meetings
- Client and stakeholder meetings
- Academic and research meetings
- Project review and planning meetings
- Software development team meetings
- Team discussions and follow-up meetings

4. Motivation for Selection of the Project

This project was selected because it addresses a gap that is both widespread and well understood: as described in Section 3.2, current meeting platforms stop at summaries and action items, leaving execution and follow-through to manual effort. In delivery organisations, where a program manager or Software Engineer may sit through several meetings a day, this gap translates directly into missed commitments, duplicated tracking effort, and decisions that lose traceability within weeks. The problem is also a natural fit for the GenAI patterns covered in this program — speech-to-text, structured extraction with hallucination controls, and tool-calling agents that act on external systems — making it a realistic test of end-to-end LLM engineering rather than a standalone model exercise.

5. Detailed Dataset / Input Data Description and Source

The current workflow is primarily designed around user-provided meeting recordings rather than a fixed dataset. The Meeting Intake Form requires a meeting title, participant email addresses, and one meeting recording. The workflow accepts MP3, M4A, WAV, MP4, and WEBM files.

For system development and testing, meeting recordings and transcripts available to the team can be used where appropriate and where permission and privacy requirements are satisfied. The current workflow itself does not contain a dataset ingestion, preprocessing, or dataset-analysis node. Therefore, a specific external benchmark dataset is not claimed as an implemented component of this workflow.

If MeetingBank or another public meeting dataset is used later for evaluation, it should be documented separately as an evaluation dataset rather than being represented as an already implemented part of the current workflow.

6. Current Benchmark

The current workflow is a functional prototype rather than a benchmark implementation. It does not currently contain a dedicated benchmark-evaluation pipeline or reference-summary comparison node. Therefore, no numerical benchmark result is claimed at this stage.

For future evaluation, the team can compare generated summaries and extracted meeting information against reference data where such references are available. Suitable evaluation can include summarization quality, action-item extraction quality, decision extraction quality, and end-to-end automation success.

- Summarization: ROUGE-based comparison when reference summaries are available.
- Action-item extraction: Precision, Recall, and F1-score against manually verified action items.
- Decision extraction: Precision, Recall, and F1-score.

- Workflow reliability: successful transcription, task creation, notification, and document-generation rates.
- Processing performance: end-to-end processing time and API response latency.

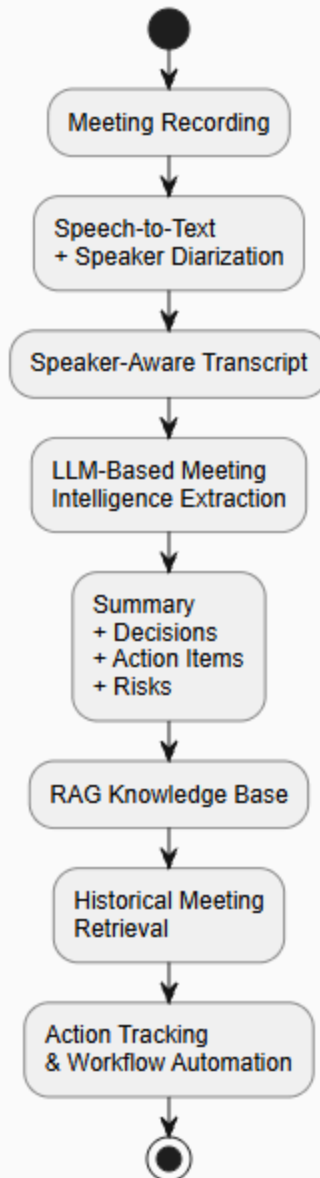
7. Proposed Plan

7.1 Methodology / Approach

The current end-to-end methodology is:

- Meeting Intake: collect meeting title, participant emails, and meeting recording.
- Audio Upload: upload the binary recording to AssemblyAI.
- Transcription: request transcription with speaker labels enabled.
- Status Polling: wait and check the transcript status until it is completed.
- Diarized Transcript Construction: convert speaker utterances into a readable speaker-wise transcript.
- Meeting Intelligence Extraction: use OpenAI to produce structured JSON containing summary, decisions, action items, risks, follow-up meeting information, and parking-lot items.
- Structured Output Assembly: normalize the extracted data and prepare email, task, and documentation outputs.
- Action Tracking: split action items and create Jira tasks.
- Documentation and Communication: create Google Docs meeting notes and email the meeting summary to participants.
- Follow-up Automation: when a follow-up is required, create a Google Calendar event.

Meeting Intelligence & Action Tracking System - Methodology



7.2 Packages and Tools

Below are the tools proposed to use

- n8n: workflow orchestration and API integration
- AssemblyAI API: audio upload, transcription, and speaker diarization
- OpenAI API: meeting-intelligence extraction and structured JSON generation
- Jira: action-item/task tracking
- Gmail: participant summary distribution
- Google Docs: meeting-note storage
- Google Calendar: follow-up meeting automation

7.3 Algorithms and Techniques

Below are the Techniques planned to use

- Speech-to-text transcription
- Speaker diarization
- Speaker-wise transcript construction
- LLM-based summarization
- Prompt-based structured information extraction
- JSON structured output
- Action-item extraction
- Rule-based normalization/default handling for missing assignees and deadlines
- Workflow branching and automation

7.4 Evaluation Metrics

The current workflow does not yet calculate evaluation metrics automatically. The following metrics are proposed for project evaluation after a verified test set is prepared:

- Automated Evaluation Metrics and LLM as a Judge or summary evaluation where reference summaries exist.
- Precision, Recall, and F1-score for action-item and decision extraction.
- Task creation success rate for Jira automation.
- Notification and document creation success rate.
- End-to-end processing time.

7.5 Stages and Defined Deliverables

Stage	Activities	Deliverables
Stage 1 – Meeting Processing	Meeting intake, audio upload, transcription, speaker diarization, status polling, diarized transcript.	Working meeting ingestion and speaker-aware transcription pipeline.
Stage 2 – Meeting Intelligence	LLM processing, summary generation, decision extraction, action-item extraction, risk and follow-up extraction, structured output.	Structured meeting-intelligence JSON and readable meeting summary.
Stage 3 – Action Tracking & Automation	Jira task creation, email distribution, Google Docs notes, follow-up Calendar event, integration testing.	End-to-end automated Meeting Intelligence & Action Tracking prototype.

7.6 Deployment Plan

The current implementation is an n8n-based workflow that can be triggered through its Meeting Intake Form. The workflow integrates external APIs and productivity platforms. For the final demonstration, the team can expose the intake interface and demonstrate the complete workflow from recording upload to generated meeting outputs.

7.7 LLMOps Plan

The current workflow provides structured LLM output and workflow-level orchestration. For the project, LLMOps practices will focus on reliable prompt execution, structured-output validation, API usage monitoring, error handling, execution logging, and evaluation of generated meeting intelligence.

- Maintain and version the extraction prompt.
- Validate structured JSON output before downstream automation.
- Monitor API failures and workflow execution failures.
- Track model usage, latency, and token consumption where available.
- Evaluate generated summaries and action items using a verified test set.
- Protect API credentials through n8n credentials rather than embedding secrets in workflow logic.

8. Preliminary Exploratory Data Analysis

The current workflow does not perform exploratory data analysis because its input is a meeting recording submitted at runtime. Therefore, no EDA results are claimed from the workflow itself.

For project evaluation, the team can perform preliminary analysis on a selected collection of meeting recordings/transcripts, examining meeting duration, transcript length, number of speakers, number of utterances, and summary length. These observations can guide processing and evaluation choices.

9. Expected Outcomes

Below are the expected outcomes

- A working system for uploading meeting recordings.
- Automatic transcription with speaker diarization.
- Structured meeting summaries and key topics.
- Automatic extraction of decisions and action items.
- Extraction of assignees, deadlines, priority, and details when stated in the meeting.
- Identification of risks and blockers.
- Automated Jira task creation for action items.
- Automatic participant email summaries.
- Automatic creation of meeting notes in Google Docs.
- Automatic follow-up calendar-event creation when a follow-up is identified.
- A demonstrable end-to-end workflow for Meeting Intelligence and Action Tracking.

10. Project Demonstration Strategy

The demonstration will use a sample meeting recording containing multiple speakers and several discussion outcomes. The team will show the workflow from input to automated outputs.

- Upload the meeting recording and enter meeting details.
- Show AssemblyAI transcription and speaker diarization.
- Show the generated speaker-aware transcript.
- Run the OpenAI meeting-intelligence extraction.
- Display summary, decisions, action items, risks, and follow-up information.
- Show the automatically created Jira tasks for extracted action items.
- Show the generated Google Docs meeting notes.
- Show the email summary sent to participants.
- Show the follow-up Google Calendar event when the workflow identifies a follow-up requirement.

11. Proposed Timeline of Project Stage Executions

Week	Stage	Weekly Goal	Expected Output
Week 1	Stage 1	Finalize architecture, meeting intake, audio upload, transcription and speaker diarization.	Working transcription pipeline.
Week 2	Stage 1 → Stage 2	Complete transcript processing and integrate structured LLM meeting-intelligence extraction.	Speaker-aware transcript + structured intelligence.
Week 3	Stage 2	Refine summaries, decisions, action items, risks and output validation; prepare evaluation set.	Validated meeting-intelligence pipeline.
Week 4	Stage 3	Integrate Jira, Gmail, Google Docs and Google Calendar; perform end-to-end testing.	Automated action tracking and collaboration workflow.
Week 5	Stage 3	Final testing, metrics, reliability checks, deployment preparation and demonstration.	Final prototype and demonstration.

12. Team Members

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13. Designated Team Coordinator

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Appendix: Current Workflow Summary

The current n8n workflow contains the following major functional groups:

- Transcribe AssemblyAI upload, transcription request, waiting, status retrieval, and completion check.
- Extract diarized transcript construction, OpenAI meeting-intelligence extraction, and structured-output assembly.
- Create Tasks split action items and create Jira/Monday.com items in the current workflow.

- Meeting Notes Doc build notes text and create/update a Google Doc.
- Follow-up Calendar check whether a follow-up is needed and create a Google Calendar event.

For the final project architecture, the team should decide whether to retain both Jira and Monday.com. Since both currently receive each extracted action item as a task, using one primary task-management platform will avoid duplicate tracking. This proposal treats Jira as the primary action-tracking destination; Monday.com is therefore not presented as a required final component.